a

Study cohort
Input: Demographics (age, sex, education) + APOE-ε4 carrier + neuropsychological data +proteomics + metabolomics;
Target: Aβ-PET status

Preprocessing
categorical encoding; min–max scaling; sex-stratified split

50 repeats

Outer loop — stratified 5-fold CV

Stratified 5-fold split

Outer training fold (4 of 5)

Held-out outer fold (1 of 5)

Inner loop — stratified 5-fold CV

Hyperparameter tuning
penalty (L1 / L2), C ∈ {0.001, 0.01, 0.1, 1, 10, 100}

Select parameters with highest mean inner-fold ROC-AUC

Refit on the full outer training fold
trained model + feature set (LR L1 model only)

Evaluate on held-out fold
ROC-AUC; accuracy, sensitivity, specificity at the Youden-index threshold

250 outer-fold evaluations (50 repeats × 5 folds)

b

